AI/ML Project

Project Title:

Personal Finance and Investment Tracker with Wants vs Needs Classification

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Student Details:

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1. Introduction

In the modern financial environment, individuals often face challenges in managing their income, expenses, and investments efficiently. Poor financial planning can lead to unnecessary spending and reduced savings. Therefore, having a structured system to monitor financial activities is essential.

This project presents a **Personal Finance and Investment Tracker** developed using Python. The system enables users to record transactions, analyze spending behavior, and monitor investment performance. Additionally, it incorporates a basic intelligent feature that classifies expenses into **wants and needs**,helping users make informed financial decisions.

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2. Problem Statement

Many individuals lack a proper system to:

- Track daily expenses
- Differentiate between essential and non-essential spending

- Monitor investment growth
- Analyze their financial health

Existing tools may be complex or require internet access and external dependencies. Hence, there is a need for a **simple, offline, and efficient financial tracking system**.

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3. Objectives

The primary objectives of this project are:

- To design a simple financial tracking system using Python
- To manage income and expenses effectively
- To classify expenses into **needs and wants**
- To track investments and calculate returns
- To compute financial indicators such as:
    - Balance
    - Net Worth
    - Profit/Loss
- To implement basic predictive logic for:
    - Future expenses
    - Investment growth
- To build the system using **core Python without external libraries**

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4. Scope of the Project

This project is intended for:

- Students managing personal expenses
- Beginners in financial planning
- Users who need a lightweight offline tool

The system operates through a **command-line interface** and does not require any additional installations, making it highly accessible.

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5. Methodology

The project follows a structured approach using fundamental programming concepts.

5.1 Data Storage

- Transactions and investments are stored using:
    - Lists
    - Dictionaries
- This eliminates the need for external libraries like pandas.

5.2 Transaction Handling

Each transaction includes:

- Date
- Type (Income/Expense)
- Amount
- Category
- Description
- Tag (Need/Want)

5.3 Wants vs Needs Classification

A rule-based approach is used:

- Keywords such as *food, rent, bills* → classified as **Needs**
- Keywords such as *shopping, movies, luxury* → classified as **Wants**

5.4 Financial Calculations

- **Balance = Total Income – Total Expenses**
- **Profit/Loss = Current Investment Value – Invested Amount**
- **Net Worth = Balance + Current Investment Value**

5.5 Prediction Logic

Instead of advanced ML models, simplified logic is used:

- Expense prediction → based on average spending
- Investment prediction → based on average growth rate

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6. System Architecture

The system consists of the following modules:

1. **Input Module**
    - Accepts user data via command line
2. **Processing Module**
    - Performs classification and calculations
3. **Storage Module**
    - Stores data in lists and dictionaries

4. **Analysis Module**
    - Generates summaries and insights
5. **Prediction Module**
    - Estimates future financial values

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7. Algorithm

7.1 Expense Classification Algorithm

1. Take category and description as input
2. Convert text to lowercase
3. Match keywords:
    - If match in needs → return NEED
    - If match in wants → return WANT
4. Else → return UNCATEGORIZED

7.2 Expense Prediction Algorithm

1. Extract all expense values
2. Calculate average
3. Return average as predicted expense

7.3 Investment Prediction Algorithm

1. Calculate growth rate for each investment
2. Find average growth rate
3. Apply it to current portfolio value

4. Return predicted value

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8. Features of the System

- Income and expense tracking
- Investment management
- Automatic wants vs needs classification
- Financial summary generation
- Net worth calculation
- Spending behavior analysis
- Simple text-based visualization
- Predictive financial insights

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9. Sample Working

Example:

- User enters:
    - Expense: Rs 500 (Zomato)
    - Classified as: WANT
- User enters:
    - Expense: Rs 2000 (Rent)
    - Classified as: NEED

Output:

- Needs Spending: Rs 2000

- Wants Spending: Rs 500
- Needs %: 80%
- Wants %: 20%

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10. Advantages

- Lightweight and fast
- No external dependencies
- Easy to understand and implement
- Useful for beginners
- Demonstrates core Python concepts
- Helps improve financial awareness

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11. Limitations

- No graphical user interface
- No data persistence (data not saved permanently)
- Basic prediction logic (not advanced ML)
- Limited scalability

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12. Future Enhancements

The system can be further improved by:

- Adding file storage (CSV/database)
- Developing a GUI using Tkinter or web frameworks

- Integrating real-time stock market APIs
- Using machine learning models for accurate predictions
- Implementing natural language processing (NLP) for smarter classification
- Adding budget alerts and notifications

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13. Applications

- Personal expense tracking
- Student financial management
- Budget planning
- Basic financial analysis tool

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14. Conclusion

This project demonstrates the development of a **personal finance and investment tracking system** using core Python. It successfully integrates financial management with basic intelligent features such as expense classification and prediction.

The system provides valuable insights into spending behavior and financial health, making it a practical tool for everyday use. Moreover, it serves as a strong foundation for implementing advanced AI/ML techniques in the future.

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15. References

- Python Official Documentation
- Basic Financial Management Principles
- Introduction to Machine Learning Concepts